

Literature Review: Positioning Athena Against the Oncology CDS Landscape

Comparative literature synthesis · oncology clinical decision support · September 2026

Source: Comparative search conducted via Consensus.app (Consensus NLP, Inc.), 8 research areas, ~40 papers screened (2018–2026), synthesized against Athena's built architecture (Concept.md, IMPLEMENTATION_PLAN.md, NEW_SUBSYSTEMS.md, Subsystems.md) and its in-progress MVP scope (issues #121–#125). Raw search exports retained at lit_review/ (repo root, gitignored).

0. Why this review, and how to read it

Athena makes four architectural bets that are each, individually, testable against a body of prior work: (1) retrieval-grounded, narrow-domain LLM synthesis over broad general-purpose querying; (2) a hard citation-verification gate rather than trusting prompt-level source markers; (3) deterministic, non-agentic decision logic (Signal Generators, Diff Engine) with LLM synthesis confined to explanation/summarization, rather than autonomous multi-agent deliberation; (4) an event-sourced, append-only, diff-first case memory where the change-review screen — not chat — is the primary clinical interface. This review checks each bet against the literature, not to justify decisions already made, but to locate exactly where Athena is executing a validated pattern versus where it is making a genuinely novel and unvalidated claim. That distinction matters for the eventual paper: reviewers will accept "we applied known-good patterns to oncology" readily; "we invented an unstudied interaction paradigm" needs its own evaluation section to survive review.

The one-line finding, stated up front: **Athena's design choices are well aligned with where the literature is strongest** — narrow-domain retrieval grounding, deterministic/hybrid control, and verification-first architecture are all independently supported. **The one component with almost no direct precedent is the event-sourced, append-only, clinician-facing diff-based case memory** — which is either Athena's weakest evidentiary foundation or its most publishable novel contribution, depending on execution.

1. Oncology-specific RAG/LLM clinical decision support

The base rate question: does retrieval-augmented generation, scoped to oncology and to a single cancer type, actually outperform general-purpose LLM querying? Six systems bear directly on this.

System / paper	Approach	Documented limitation
MEREDITH (Lammert et al., 2024, <i>JCO Precision Oncology</i>)	Gemini Pro + RAG + chain-of-thought over PubMed, trial data, drug approval status, guidelines; refined against molecular tumor board (MTB) feedback	High concordance with MTB recommendations (94.7%), but evaluated on only 10 fictional cases and partly on a proprietary institutional-access corpus
Context-augmented precision oncology LLM (Jun et al., 2026, <i>Cancer Cell</i>)	RAG-LLM using the MOAImanac database + structured augmentation for biomarker-driven therapy retrieval	Strong synthetic (95%) and real-world (93%) accuracy, but authors frame it as deployment guidance rather than proof of clinical safety or outcome benefit
Breast-cancer open-source RAG benchmark (Park et al., 2026, <i>JCO Clinical Cancer Informatics</i>)	HTML-structure-preserving chunking over 1,356 ASCO breast guideline documents; tested GPT-4-turbo, GPT-3.5, Qwen2.5-14B, LLaMA3-8B, OpenBioLLM-8B with/without RAG	Small open-source models with RAG approached but did not match GPT-4 (rubric 3.77 vs 3.96); performance degraded sharply past top-5 retrieved contexts; human raters were more conservative than LLM-judge scores
Guideline-anchored gynecologic oncology benchmark (Dukes et al., 2026, <i>Gynecologic Oncology</i>)	GPT-5 + NCCN-anchored RAG vs. baseline GPT-5 and literature-only RAG	Guideline-anchored RAG outperformed both baselines, but the study is benchmark-only — no clinical safety or outcome evidence
DLBCL oncology RAG model (Soong et al., 2023, <i>PLOS Digital Health</i>)	Domain-specific RAG answering lymphoma biology/treatment questions vs. GPT-3.5/4 and Microsoft Prometheus	RAG model had higher accuracy/relevance; GPT-3.5/4 produced more nonexistent references and inaccurate answers, but the evaluation set was only 19 questions

Where Athena stands. This is the strongest-supported bet in the whole architecture. Every paper in this area finds the same shape of result: retrieval grounded in a curated, domain-scoped corpus beats bare LLM querying, and the effect holds even for small open-weight models once retrieval is tight (Park et al., 2026) — directly validating both the single-cancer-type MVP decision (issue #121) and the open-source LLM plan (issue #122). The recurring caveat across all five studies — small case counts, benchmark-only evaluation, no prospective clinical outcome data — is not a criticism unique to Athena; it is the state of the entire field, and it means Athena's own eventual evaluation (the pre-committed gold set required by #121) would already be more rigorous than most published comparators if it includes real clinician review rather than LLM-judge scoring alone.

2. Citation hallucination and source-attribution failure

This is the literature base for the Citation Verification Gate (issue #9, extended by #124).

System / paper	Approach	Documented limitation
McGowan et al., 2023, <i>Psychiatry Research</i>	Tested ChatGPT and Bard on citation generation for psychiatry literature search	Of 35 generated citations, only 2 were real in one test run — "spontaneous citation fabrication"
Chelli et al., 2024, <i>Journal of Medical Internet Research</i>	Compared GPT-3.5, GPT-4, Bard against 11 human-conducted systematic reviews (471 references)	Precision 9.4–13.4% (0% for Bard); hallucination rates 28.6–91.4%; concluded LLMs are not recommendable as primary systematic-review tools
Linardon et al., 2025, <i>JMIR Mental Health</i>	GPT-4o generating literature reviews across topics of varying visibility/specialization	19.9% of 176 citations fully fabricated, 45.4% of "real" citations contained errors; fabrication risk rose specifically for less visible, more specialized topics
Gupta et al., 2024 (TREC 2024 BioGen Track), <i>arXiv</i>	Structured benchmark track for biomedical reference attribution	LLMs "often fail to reference relevant sources," especially on lay-user-phrased questions — unsupported statements framed as the primary barrier to health-domain LLM use
Med-V1 (Jin et al., 2026), <i>arXiv</i>	3B-parameter model purpose-built for zero-shot biomedical evidence attribution/verification	Comparable to GPT-5 on attribution tasks at much lower cost, but citation validity remains sensitive to output-format instructions, and misattribution still occurs in guideline-adjacent high-stakes text

Where Athena stands. The literature's finding that fabrication risk rises specifically in "less visible or specialized" topics (Linardon et al., 2025) is the single most load-bearing citation for Athena's whole premise: rare cancers, uncommon biomarkers, and off-label access questions are exactly that regime. This directly strengthens the argument already made in issue #124 — that #107/#108's SOURCE_MARKER/CONTEXT_MARKER fix is a prompt-level hint a real generative model can still ignore, and that a structural, enforced post-hoc check (`citation_gate.py` verifying against `Turn.sources`, not just trusting the prompt) is the architecturally correct response, consistent with what Gupta et al. (2024) and Jin et al. (2026) both converge on: verification has to be a separate, checkable step, not a property hoped for from prompting alone.

3. Multi-agent / "AI tumor board" systems

This is the literature that bears directly on the tumor-board decision already made this session (T–AF marked non-blocking, #73 comment, subsystems T–AF tracked as background research only).

System / paper	Approach	Documented limitation
EvoMDT (Liu et al., 2026, <i>NPJ Digital Medicine</i>)	Self-evolving multi-agent system: domain agents + consensus protocol + retrieval scope that updates from expert feedback	Comparable decision quality to human MDTs, 30–40% faster, but authors themselves note it "remains a complex architecture whose real-world governance burden is part of the problem it tries to solve"
Tumor-board multi-agent LLM for HCC (Saenz et al., 2026, <i>Journal of Clinical Oncology</i>)	Three specialist personas (hepatology/oncology/radiology) + supervisor agent, RAG over AASLD/EASL/ESMO/BSG guidelines	Accuracy improved over single-call inference (+6.1 to +9.4 pts), but prospective validation on efficiency, guideline adherence, and patient outcomes is explicitly unstudied
MDAT gynecologic oncology (Kuerbanjiang et al., 2025, preprint)	Fixed-size multidisciplinary agent teams for complex cases	Structured consensus beat adversarial prompting, but prompt-strategy choice materially changed rankings — benchmarking was intentional, limiting deployment conclusions
Oncology MTB preparation study (Li et al., 2026, <i>CHI</i>)	Compared off-the-shelf assistant vs. task-specific multi-agent system, with oncologists in the loop	Oncologists preferred the task-specific system, but source links and agent reasoning trajectories failed to calibrate trust — summaries "still induced overconfidence"
OncoAgents + OncoGraph (Loaiza-Bonilla et al., 2026, <i>ESMO Real World Data and Digital Oncology</i>)	Neuro-symbolic multi-agent trial matching with deterministic graph checks, 3,804-patient prospective evaluation	Strong prospective results, but the benefit is attributed largely to separating LLM extraction from deterministic eligibility logic — not to free-form multi-agent deliberation itself

Where Athena stands. This is the section that most directly vindicates the reversal made mid-session this project: the goal restated by the user ("only judgment and experience should remain the doctor's job — the system should remove memory burden") is in real tension with a multi-persona debate system, and the literature agrees for a different reason. Every gain reported for multi-agent oncology systems is attributed by the authors themselves to structure imposed *around* the agents — guideline retrieval, deterministic eligibility graphs, consensus protocols with fixed weights — not to the free-form multi-persona debate itself (most explicit in Loaiza-Bonilla et al., 2026, and echoed by EvoMDT's own authors). Li et al. (2026) is the sharpest warning: even when a multi-agent system performs well, exposing its reasoning trajectory to clinicians did not fix overconfidence — it can still mislead calibrated trust. This is the literature-grounded version of the argument already made informally: Athena's deterministic Signal Generator Layer (#6) plus Access Pathway Registry (#11) is functionally "the part of a tumor board that works" (structured, retrieval-grounded, auditable), without the part that doesn't (autonomous persona debate as the primary decision mechanism).

4. Deterministic vs. agentic clinical decision support

The architecture-level question underneath §3: when should an LLM's output be allowed to directly drive a clinical decision, versus constrained by a deterministic layer?

System / paper	Approach	Documented limitation
KRD (Zhang & Chen, 2026, <i>Journal of Intelligent Medicine and Healthcare</i>)	Fact extraction + compiled knowledge + handwritten rule layer + decision interface, short-circuiting on hard safety violations	Reached the "hard-safety ceiling" with better evidence-trace completeness and lower reviewer burden than lighter hybrids, but the evidence gate was inert on this benchmark and failures clustered in the extraction step
AnemiaCare HD (Arriola-Montenegro et al., 2025, <i>Frontiers in AI</i>)	Deterministic prompt logic for protocol-bound anemia management	Loosely structured prompting reached only 32% protocol adherence; deterministic encoding reached 100% on simulated cases — but real-world/prospective validation is still missing
CPGPrompt (Deng et al., 2026, <i>JAMIA</i>)	Guideline narratives translated into decision trees traversed by structured yes/no LLM queries, fully logged	Binary referral decisions were strong; multiclass pathway assignment dropped, with negation and temporal reasoning as the key failure modes
Medi-Gemma (Quadri et al., 2026, <i>arXiv</i>)	Deterministic EMR analytics + RAG + intent routing + ground-truth-state injection + safety verifier	Built specifically to fix temporal inconsistency and structured-data hallucination in conventional RAG, but prospective multi-site validation remains future work
Neuro-symbolic report extraction (Prenosil et al., 2025, <i>Communications Medicine</i>)	GPT-4 candidate extraction + rule-based expert-system verification	Hybrid system matched physicians and intercepted residual identifiers — supports deterministic verification layers for sensitive workflows

Where Athena stands. The consistent finding — a deterministic layer reaching near-100% adherence where loose LLM prompting reaches ~32% (Arriola-Montenegro et al., 2025) — is the direct empirical justification for keeping the Diff Engine (#5) and Signal Generator Layer (#6) fully deterministic rather than migrating any of that logic into LLM judgment calls, which nothing in this session's design has proposed but which is worth stating explicitly as a design constraint going forward. The recurring failure mode across CPGPrompt and KRD — errors clustering in extraction/negation/temporal reasoning, not in the deterministic core — also directly validates why Automated Case Intake (#19) treats extraction as strictly human-confirmed rather than auto-committed, and reinforces that the logged-but-not-built raw-document de-identification gap in #19 sits exactly at this failure-prone extraction boundary.

5. Event-sourced / diff-based clinical records — the sparse area

This is the literature underneath Case Memory (#4) and the Diff Engine (#5), and it is, honestly, thin.

System / paper	Approach	Documented limitation
MedBeads (Nakajima, 2026, preprint)	Immutable "beads," patient-rooted Merkle DAG, reconstructable clinical links, append-only pods	Closest conceptual match to Athena found in this search; engineering feasibility tested only on synthetic data, no test of hallucination reduction or clinical outcomes
Noteless collaborative EMR (Steinkamp et al., 2020, <i>JMIR Formative Research</i>)	Single dynamic workspace organized by clinical problem, with granular version history	Demonstrates feasibility of non-note longitudinal state and change tracking, but is not an explicit append-only event-sourced architecture
Event-sourced systems schema-evolution study (Overeem et al., 2021, <i>Journal of Systems and Software</i>)	Cross-industry empirical study of 19 event-sourced systems	Core challenges: evolution, steep learning curve, tooling, rebuilding projections, privacy — but this is general systems literature, not clinician-facing oncology CDS
Baseline (Edwards & Petříček, 2025, <i>arXiv</i>)	Operation-based data version control with high-fidelity differencing across schema changes	Relevant for diff fidelity and schema evolution generally, but not clinical, and not focused on bedside review workflows
CQRS/event-sourcing architecture studies (Palamakula, 2026; Lytvynov & Hruzín, 2024/2025/2026)	Compliance-oriented and snapshot-centric CQRS/ES variants for audit and regulatory reporting	Articulate audit/replay trade-offs, but emphasize engineering complexity, causal ordering, and performance-maintenance trade-offs rather than clinician-facing diff UX

Where Athena stands — and this is the finding to lead the paper's contribution claim with. There is essentially no literature on a clinician-facing interface where the *primary interaction loop* is reviewing a computed diff over patient state and evidence, as opposed to a chronological note timeline (Steinkamp et al.) or a backend audit log (Overeem et al.; Lytvynov & Hruzín). MedBeads (Nakajima, 2026) is the nearest analog and remains untested on real clinical outcomes or hallucination reduction. This means the design law already written into `RESEARCH_DIRECTION.md` — "chat hides the diff; the diff is the product" — is not validated prior art being applied to oncology; it is close to a novel interaction paradigm for this domain. That cuts both ways for the eventual paper: it is the strongest available novelty claim, but it also means Athena cannot cite precedent for "diff-first review improves clinician error detection" — that claim has to be earned by Athena's own evaluation, not borrowed from the literature. This should be treated as a first-class open research question in the paper, not a background assumption.

6. Open-source / self-hostable medical LLMs

Direct evidentiary base for issue #122.

System / paper	Approach	Documented limitation
BioMistral (Labrak et al., 2024, preprint)	Open-source biomedical pretraining on PMC full text, multilingual benchmark release	Strong among open medical models, but frontier proprietary models still lead on many open-ended and reasoning tasks
MEDITRON-70B (Chen et al., 2023, arXiv)	Open 7B/70B Llama-2 adaptation with guidelines + PubMed pretraining	Competitive with GPT-4-class models but stays within ~5% of GPT-4 and ~10% of Med-PaLM 2 rather than surpassing them
PMC-LLaMA (Wu et al., 2024, JAMIA)	Open-source medical adaptation using 4.8M papers, textbooks, instruction tuning	Strong QA benchmark results, but benchmark gains don't guarantee oncology workflow safety or current-knowledge performance
Fully Open Meditron (Theimer-Lienhard et al., 2026, arXiv)	Fully open end-to-end pipeline, clinician-audited corpus, reproducible training	Fixes open-weight opacity, but is early-stage and mostly benchmark-based rather than evidence of oncology-specific deployment
Cross-model oncology benchmark studies (Longwell et al., 2024, JAMA Network Open; Warner et al., 2026, arXiv)	Cross-model benchmarks on oncology board-exam-style questions and clinical reasoning	Best proprietary models still outperform best open models on oncology exams; incorrect oncology answers carry "moderate-to-severe harm potential"

Where Athena stands. The pattern here is a genuine trade-off, not a clean win: open-weight biomedical models (BioMistral, MEDITRON) are close enough to frontier proprietary performance to be viable for privacy-preserving, self-hosted deployment (directly supporting the LLMClient Protocol's model-agnostic design in `llm_client.py`), but "close" is not "equal" — Longwell et al. (2024) and Warner et al. (2026) both find open models still trail on oncology-specific reasoning, and incorrect answers there carry real harm potential. This is the strongest argument in the whole review for why Athena's open-source LLM should sit *behind* the Citation Verification Gate and Diff Engine rather than generate unconstrained clinical text — the deterministic scaffolding is not just an architectural preference, it is compensating for a documented, unresolved capability gap in the exact model class Athena plans to deploy.

7. Chat-first vs. structured-review-first clinical interfaces

Human-factors evidence for the App.jsx routing fix (issue #123) and for keeping chat secondary to the change-review screen generally.

System / paper	Approach	Documented limitation
GutGPT / LLM-augmented CDSS trial (Rajashekar et al., 2024, <i>CHI</i> ; Chan et al., 2023, <i>arXiv</i>)	Compared dashboard+LLM vs. dashboard+search in simulation	Improved ease-of-use, sometimes improved trust/mastery, but acceptance effects were mixed, not uniformly positive
Conversational XAI vs. dashboard (He et al., 2025, <i>IUI</i>)	Compared conversational explanation interfaces with dashboards	Understanding and trust improved, but over-reliance remained, and LLM-agent enhancements specifically <i>amplified</i> over-reliance
Automation-bias literature review (Romeo & Conti, 2025, <i>AI & Society</i> ; Jovchevski et al., 2026, <i>Philosophy & Technology</i>)	Synthesis of overreliance experiments in AI-supported decisions	Explanations alone often fail to improve accuracy or mitigate automation bias — increased verification effort and "epistemic friction" are more promising interventions
Clinician trust / copilot studies (Stevens & Stetson, 2023, <i>JB</i> ; Zhu et al., 2026, <i>CHI</i>)	Trust instruments + interactive, visualization-first copilot studies	Trust is built through verification and interpretable visualization, favoring interfaces that support adversarial checking over passive acceptance
Direct automation-bias trial with physicians (Qazi et al., 2025)	Randomized physicians to correct vs. flawed ChatGPT-4o recommendations	Erroneous chat-style recommendations reduced diagnostic reasoning accuracy by 14 points, <i>even in AI-trained physicians</i>

Where Athena stands. This is the direct evidentiary backing for issue #123, and it raises the stakes on it: Qazi et al. (2025)'s 14-point accuracy drop from erroneous chat-style advice, in physicians already trained on AI tools, is not a hypothetical risk — it is close to a description of what a chat-first landing page invites. He et al. (2025)'s finding that LLM-agent enhancements specifically *amplify* over-reliance (rather than being neutral) sharpens the case further: Athena's chat surface is exactly the kind of enhancement that risks that effect if it is the first thing a clinician sees, rather than a secondary, citation-gated layer reached from within a case's diff review. The literature's proposed mitigation — "epistemic friction" and interfaces built for adversarial verification rather than passive acceptance (Romeo & Conti, 2025; Stevens & Stetson, 2023) — is a good design lens to apply to the change-review screen's actual layout, not just its route priority.

8. Trial matching and access-pathway systems

Evidentiary base for the Access Pathway Registry (#11).

System / paper	Approach	Documented limitation
TrialGPT (Jin et al., 2023, <i>Nature Communications</i>)	End-to-end LLM framework for patient-to-trial retrieval, criterion matching, ranking	Strong retrieval/ranking on synthetic cohorts with real screening-time reduction, but no prospective real-world oncology deployment in the original evaluation
TrialMatchAI (Abdallah et al., 2025, <i>Nature Communications</i>)	Open-source RAG + fine-tuned open LLMs + criterion-level eligibility reasoning	92% of oncology patients had a relevant trial in the top 20, >90% criterion-level accuracy — but still requires expert assessment, and focuses on trials more than expanded-access pathways specifically
Australian lung-cancer trial matching (Alexander et al., 2020, <i>JAMIA Open</i>)	AI eligibility matching against clinician gold standard	Efficient and accurate prescreening, but clinician oversight remained necessary throughout
OncoAgents/OncoGraph prospective platform (Loaiza-Bonilla et al., 2026, <i>ESMO Real World Data and Digital Oncology</i>)	Multi-agent extraction + deterministic oncology knowledge graph, 3,804 patients	High F1 and efficiency gains, but strength attributed to neuro-symbolic constraint + clinician review, not autonomous LLM recommendation
Off-label/expanded-access frameworks: BELIEVE (Ishimaru et al., 2023, <i>Int. J. Clinical Oncology</i>), DRUP-style genomics-guided off-label (Verkerk et al., 2026, <i>Nature</i>), MTB frameworks (Rolfo et al., 2018, <i>ESMO Open</i>)	Structured, protocolized programs for post-standard-option access	Address the "no standard options remain" problem directly, but benefit is modest overall and works best in systematic programs, not ad hoc off-label use

Where Athena stands. This is the section that most directly validates a specific, previously under-examined part of Athena's design: the *access-pathway* half of #11 (as distinct from ordinary trial matching) has real precedent in BELIEVE, DRUP, and MTB-framework literature, but that precedent consistently warns that benefit is "modest overall" and works best inside systematic, protocolized programs — not as an ad hoc recommendation surfaced by a general system. This is a meaningful scoping signal: Athena's Access Pathway Registry should be evaluated (and probably described in the eventual paper) as a *structured, protocol-anchored* pathway-matching tool in the lineage of BELIEVE/DRUP, not as a generic "find me options" feature — the literature's caution about ad hoc off-label suggestion applies directly to how permissive this subsystem should be by default.

9. Cross-cutting synthesis

Three patterns recur across all eight sections independently of which subsystem they bear on:

- 1. Retrieval grounding, scoped narrowly, is the single most consistent source of measured improvement** across oncology LLM systems (§1, §6, §8) — and the effect holds even for small open-weight models once the retrieval corpus is tightly curated (Park et al., 2026), which is the strongest available justification for the MVP's single-cancer-type decision over broad multi-cancer coverage.
- 2. No oncology system in this review relies on a bare LLM.** Every system that reports strong results combines retrieval with either structured metadata, ontology normalization, or

deterministic logic that constrains what the model is permitted to say or do (Lammert et al., 2024; Loaiza-Bonilla et al., 2026; Quadri et al., 2026; Deng et al., 2026). Athena's Signal Generators, Diff Engine, and Citation Verification Gate are this exact pattern, applied consistently rather than partially.

3. Citation fabrication risk is not a solved problem and is worse exactly where

Athena's use case lives — specialized, lower-visibility topics (Linardon et al., 2025), and the literature's own preferred mitigation is combined retrieval-plus-verification-plus-human-review (Basu & Huynh, 2026), not prompt-level fixes alone — directly supporting #124's argument that #107/#108's marker-based fix needs a structural, enforced backstop.

10. Design implications already reflected in this session's decisions

Two decisions made earlier in this project's issue-restructuring work turn out to have direct literature support that wasn't available at decision time:

- **Rejecting a multi-agent "AI tumor board" as the primary shipped mechanism** (tracked as non-blocking background research in #73, #75–#87) is supported, not just by the internal tension the user identified between a multi-persona board and the "remove memory burden, keep only judgment" goal, but independently by the literature: every multi-agent oncology system's measured gains are attributed by its own authors to added structure (retrieval, graphs, consensus protocols) rather than to free-form deliberation itself (§3), and at least one study found multi-agent reasoning trajectories failed to correct clinician overconfidence even when exposed (Li et al., 2026).
- **Making the change-review/diff screen the primary route, chat secondary** (issue #123) is supported by direct evidence that chat-style advice measurably degrades physician diagnostic accuracy when wrong (Qazi et al., 2025: –14 points, even in AI-trained physicians) and that LLM-agent-style interfaces specifically amplify over-reliance relative to plainer dashboards (He et al., 2025) — this is not just an internal design-law violation (RESEARCH_DIRECTION.md), it is a shipped configuration the literature would flag as a measurable risk factor.

11. Gaps and positioning — what to claim in the paper

The single sparsest area, and the one to lead the contribution section with, is §5: event-sourced, append-only, clinician-facing diff-based case memory. Adjacent literature exists (immutable clinical context graphs, general event-sourcing/CQRS architectures, non-note longitudinal EMR designs) but essentially nothing directly studies a clinical decision-support interface where reviewing a computed diff over patient state and evidence is the primary interaction loop. That sparsity should be treated as the paper's actual novelty claim — not "we built an oncology LLM system" (crowded, §1/§6/§8 show dozens of comparable projects), but "we applied a diff-first, event-sourced review paradigm — validated in software engineering but never in clinician-facing CDS — to oncology, with citation-gated LLM synthesis and deterministic decision logic behind it."

The honest caveat that belongs in the paper's limitations section: because §5 is sparse, Athena cannot cite precedent for its central UX claim ("diff-first review improves clinician error detection or trust calibration versus chat or dashboard review") the way it can for retrieval grounding or citation verification. That claim needs to be an explicit, first-class hypothesis tested in Athena's own evaluation (the pre-committed gold set required by #121), not an assumption inherited from related work. Everything else in the architecture — narrow retrieval, deterministic control, hard citation verification, structured access-pathway matching, open-weight models behind a verification gate — has direct, multi-paper support and can be cited as applying established best practice to a new domain rather than needing to be independently proven from first principles.

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